

Cubic Boolean Functions with Small Support on Ordinary and Rainbow Johnson Slices

Runfa Hu

October 2026

1 Preliminaries and Notation

We study functions taking only the values 0 and 1 on collections of sets of a fixed size. The question is how few sets can receive the value 1 when the function is required to belong to a specified real linear subspace. We first define the collections of sets and the permitted functions, and then state the question precisely.

For a positive integer n , write $[n] = \{1, \dots, n\}$. For a finite set V , let $\binom{V}{j}$ denote the collection of its j -element subsets. For $0 \leq k \leq n$, the collection $\binom{[n]}{k}$ is called the ordinary k -slice.

We also study collections in which the underlying elements are divided into classes and a set may contain at most one element from each class. To describe this restriction, let m, a be positive integers and partition a set S into m disjoint classes of size a :

$$S = C_1 \dot{\cup} \dots \dot{\cup} C_m, \quad |C_i| = a \quad (i \in [m]).$$

We call the C_i color classes and their elements symbols. The set S with this partition is a balanced colored ground set; balanced means that all color classes have the same size. Thus m counts color classes, whereas $|S| = ma$ counts symbols. A subset of S is called rainbow if it contains at most one symbol from each color class. For $0 \leq j \leq m$, the collection of rainbow j -element subsets is

$$\mathcal{R}_j(S) = \{X \subseteq S : |X| = j, |X \cap C_i| \leq 1 \text{ for every } i \in [m]\}.$$

We call $\mathcal{R}_j(S)$ the rainbow j -slice.

Fix an integer k with $0 \leq k \leq m$. The functions considered below have domain $\mathcal{R}_k(S)$. To form a member of $\mathcal{R}_k(S)$, choose k color classes and one symbol from each chosen class. Hence

$$|\mathcal{R}_k(S)| = \binom{m}{k} a^k.$$

When $a = 1$, label the single symbol in C_i by i . Under this labeling, $\mathcal{R}_k(S)$ is the ordinary slice $\binom{[m]}{k}$.

To express the restriction on functions using linear algebra, let $\text{Fun}_k(S)$ be the real vector space of functions $f : \mathcal{R}_k(S) \rightarrow \mathbb{R}$, with pointwise addition and scalar multiplication. A function $f \in \text{Fun}_k(S)$ is Boolean if $f(X) \in \{0, 1\}$ for every $X \in \mathcal{R}_k(S)$. Write $\text{Bool}_k(S)$ for the set of Boolean functions on $\mathcal{R}_k(S)$.

For a Boolean function, we want to count the sets on which its value is 1. More generally, for any $f \in \text{Fun}_k(S)$, define its support to be the collection of sets on which its value is nonzero:

$$\text{supp } f = \{X \in \mathcal{R}_k(S) : f(X) \neq 0\}.$$

If f is Boolean, each member of $\text{supp } f$ contributes exactly 1 to the sum of its values. Therefore

$$|\text{supp } f| = \sum_{X \in \mathcal{R}_k(S)} f(X).$$

A simple Boolean function tests whether a set contains a prescribed subset. Fix $0 \leq t \leq k$ and a rainbow t -set T . The collection of sets in $\mathcal{R}_k(S)$ containing T is

$$\mathcal{S}_T = \{X \in \mathcal{R}_k(S) : T \subseteq X\}.$$

We call $\mathcal{S}_T$ the rainbow t -star with center T . Its indicator function assigns 1 to members of $\mathcal{S}_T$ and 0 to all other sets in $\mathcal{R}_k(S)$:

$$b_T(X) = \begin{cases} 1, & T \subseteq X, \\ 0, & T \not\subseteq X. \end{cases}$$

In particular, $\text{supp } b_T = \mathcal{S}_T$. To count the members of $\mathcal{S}_T$, extend T to a rainbow k -set by choosing $k - t$ of the $m - t$ colors not used by T and one symbol from each chosen color. Thus every rainbow t -star has size

$$\sigma_{m,k,t}^{\text{rb}}(a) := \binom{m-t}{k-t} a^{k-t}. \quad (1)$$

When $a = 1$, the family $\mathcal{S}_T$ is an ordinary t -star, with size $\binom{m-t}{k-t}$.

We now specify the permitted functions: they are real linear combinations of the indicators b_T whose centers have a fixed size t . For $0 \leq t \leq k$, define

$$U_t^S(k) = \text{span}_{\mathbb{R}}\{b_T : T \in \mathcal{R}_t(S)\}.$$

Equivalently, $f \in U_t^S(k)$ if there are real coefficients c_T , indexed by rainbow t -sets, such

that

$$f(X) = \sum_{T \in \binom{X}{t}} c_T \quad (X \in \mathcal{R}_k(S)). \quad (2)$$

Indeed, in a linear combination $\sum_T c_T b_T(X)$, only the terms with $T \subseteq X$ contribute. The coefficients c_T may be negative. Requiring f to be Boolean means that the sum in (2) belongs to $\{0, 1\}$ for every $X \in \mathcal{R}_k(S)$; the individual coefficients are not required to belong to $\{0, 1\}$.

To compare the spaces $U_t^S(k)$ for different values of t , keep S and k fixed and let $0 \leq r \leq t \leq k$. We claim that $U_r^S(k) \subseteq U_t^S(k)$. It suffices to express each rainbow r -star indicator as a linear combination of rainbow t -star indicators. For a rainbow r -set D , the required identity is

$$b_D = \frac{1}{\binom{k-r}{t-r}} \sum_{\substack{T \in \mathcal{R}_t(S) \\ D \subseteq T}} b_T. \quad (3)$$

Indeed, fix $X \in \mathcal{R}_k(S)$. If $D \subseteq X$, then $b_T(X) = 1$ precisely when $D \subseteq T \subseteq X$. There are $\binom{k-r}{t-r}$ such t -sets, so the right-hand side of (3) equals $1 = b_D(X)$. If $D \not\subseteq X$, then $b_D(X) = 0$ and every term on the right-hand side is also 0. This proves (3), and hence the claimed inclusion.

Consequently, for fixed S and k ,

$$U_0^S(k) \subseteq U_1^S(k) \subseteq \cdots \subseteq U_k^S(k) = \text{Fun}_k(S).$$

The space $U_0^S(k)$ consists of constant functions because $b_\emptyset = 1$. The equality $U_k^S(k) = \text{Fun}_k(S)$ holds because, for each $T \in \mathcal{R}_k(S)$, the function b_T equals 1 at T and 0 at every other member of $\mathcal{R}_k(S)$; these functions form a basis of $\text{Fun}_k(S)$.

To record the smallest center size needed to represent a function, define the Johnson degree of a nonzero $f \in \text{Fun}_k(S)$ by

$$\deg_J(f) = \min\{t \in \{0, \dots, k\} : f \in U_t^S(k)\},$$

and set $\deg_J(0) = -1$. Since $U_r^S(k) \subseteq U_t^S(k)$ for $r \leq t$, the condition $\deg_J(f) \leq t$ is equivalent to $f \in U_t^S(k)$. In the ordinary case $a = 1$, we write $U_t(k)$ for $U_t^S(k)$ on $\binom{[m]}{k}$.

We can also describe $U_t^S(k)$ using polynomials, which gives a direct way to check whether an explicit function belongs to this space. For each symbol $u \in S$, define $x_u(X) = 1$ if $u \in X$ and $x_u(X) = 0$ otherwise. Then

$$b_T(X) = \prod_{u \in T} x_u(X).$$

Thus every function in $U_t^S(k)$ has a polynomial representation of total degree at most t in the variables x_u .

Conversely, consider a polynomial of total degree at most t in the variables x_u . On $\mathcal{R}_k(S)$, the identities $x_u^2 = x_u$ remove repeated powers, and a monomial containing two distinct symbols from one color class vanishes. Each remaining monomial is therefore b_D for a rainbow set D with $|D| \leq t$. Identity (3) places every such b_D in $U_t^S(k)$, so the polynomial restricted to $\mathcal{R}_k(S)$ belongs to $U_t^S(k)$. Hence membership in $U_t^S(k)$ is equivalent to having a polynomial representation of degree at most t on $\mathcal{R}_k(S)$. A cubic Boolean function means a Boolean function of Johnson degree at most 3.

For fixed S , k , and t , our question is how small $|\text{supp } f|$ can be for a nonzero Boolean function $f \in U_t^S(k)$. Every rainbow t -star indicator b_T is such a function and has support size $\sigma_{m,k,t}^{\text{rb}}(a)$. Thus the minimum possible support size is at most $\sigma_{m,k,t}^{\text{rb}}(a)$. We ask whether the minimum equals $\sigma_{m,k,t}^{\text{rb}}(a)$, and whether every function attaining this size must be a star indicator b_T . For the parameters below, Wu conjectures that both assertions hold.

Conjecture (Wu, Conjecture 10 [1]). Let $t \geq 2$ and $k \geq t$ be integers, and put

$$q = k - t + 1, \quad m = q^2 + q + t - 1.$$

Assume $q > 2^{t-1}$, let a be a positive integer, and let S have m color classes, each of size a . If

$$f \in \text{Bool}_k(S) \cap U_t^S(k), \quad 0 < |\text{supp } f| \leq \sigma_{m,k,t}^{\text{rb}}(a),$$

then there is a fixed rainbow t -set T such that

$$f(X) = b_T(X) \quad \text{for every } X \in \mathcal{R}_k(S).$$

2 Overview of the Main Results

Wu proves Conjecture 10 for $t = 2$ [1, Theorem 11]. More explicitly, when $k \geq 4$, $m = k^2 - k + 1$, and all m color classes have the same size $a \geq 1$, every nonzero Boolean function in $U_2^S(k)$ has support at least $\sigma_{m,k,2}^{\text{rb}}(a)$, with equality precisely for rainbow 2-star indicators. For each fixed t , he also proves that every nonzero Boolean function in $U_t(k)$ on $\binom{[n]}{k}$ has support at least $\binom{n-t}{k-t}$, with equality precisely for t -star indicators, provided that k and n/k are sufficiently large in terms of t [1, Theorem 8]. For $t = 3$, Wu reports that Conjecture 10 holds for every $q \geq 5$ and $a \geq 1$; he attributes this cubic result to a separate manuscript in preparation [1, Section 7.1].

Our aim is to determine whether the same support bound and star classification hold for smaller values of q . In Section 7.1 of [1], Wu explains that the condition $q > 2^{t-1}$ is required by his method and asks whether the conclusion remains valid at $q = 2^{t-1}$. For

$t = 3$, the parameter relations become

$$k = q + 2, \quad m = q^2 + q + 2.$$

The conjecture's condition is therefore $q > 4$, and its boundary is $q = 4$.

In Problem 32 of [1], Wu asks whether the conclusion remains valid at $t = 3$ and $q = 4$ for every $a \geq 1$. The ordinary specialization $a = 1$ has $(m, k) = (22, 6)$ and star size $\binom{19}{3} = 969$. For a nonzero Boolean $f \in U_3(6)$ on $\binom{[22]}{6}$ with $|\text{supp } f| \leq 969$, Wu already shows that $|\text{supp } f| = 969$ and that two distinct members of $\text{supp } f$ cannot meet in 0 or 2 elements. The remaining question in this ordinary case is whether f must be a 3-star indicator. We complete this classification and also examine $q = 3$ and $q = 2$. All three values lie outside the conjecture's stated range $q > 4$.

Theorem 2.1 (Failure of the extension at $t = 3$, $q = 2$, and $a = 1$). *There exists a Boolean function $f \in U_3(4)$ on $\binom{[8]}{4}$ such that*

$$|\text{supp } f| = \binom{5}{1} = 5,$$

but $f \neq b_T$ for every $T \in \binom{[8]}{3}$.

Thus the proposed extension to $q = 2$ fails already in the ordinary case: a Boolean function can have the same support size as a 3-star without being a star indicator.

Theorem 2.2 (The star conclusion at $t = 3$ and $q = 3$). *Let a be a positive integer, and let S have 14 color classes, each of size a . Every nonzero Boolean function $f \in U_3^S(5)$ satisfies*

$$|\text{supp } f| \geq \binom{11}{2} a^2 = 55a^2.$$

Equality holds if and only if $f = b_T$ for a rainbow triple $T \in \mathcal{R}_3(S)$.

This theorem extends both the support bound and the star classification to $q = 3$ for every $a \geq 1$. Its ordinary specialization gives minimum support 55 on $\binom{[14]}{5}$, attained precisely by 3-star indicators.

Theorem 2.3 (The star conclusion at $t = 3$ and $q = 4$). *Let a be a positive integer, and let S have 22 color classes, each of size a . Every nonzero Boolean function $f \in U_3^S(6)$ satisfies*

$$|\text{supp } f| \geq \binom{19}{3} a^3 = 969a^3.$$

Equality holds if and only if $f = b_T$ for a rainbow triple $T \in \mathcal{R}_3(S)$.

Theorem 2.3 answers Problem 32 affirmatively for every $a \geq 1$. Together with Wu's reported cubic result for $q \geq 5$ [1, Section 7.1], Theorems 2.2 and 2.3 establish the star

support bound and equality classification for $t = 3$, every $q \geq 3$, and every balanced color-class size $a \geq 1$.

Theorem 2.1 shows that the star classification fails at $q = 2$ already when $a = 1$: a non-star Boolean function can have the same support size as a 3-star. Thus the support bound and star classification cannot both hold for every $q \geq 2$ and every $a \geq 1$.

3 Tools and Perspectives for the Proofs

We first show that, for the parameters needed below, the coefficients in a function's star-indicator representation are uniquely determined by its values. We then express the minimum-support question as a question about vectors with entries zero and one in the image of a matrix.

3.1 Full column rank of inclusion matrices

Let V be a set of size v , and let $0 \leq t \leq k \leq v$. To record which t -sets are contained in which k -sets, define the inclusion matrix $W_{v,k,t}$ with rows indexed by $X \in \binom{V}{k}$, columns indexed by $T \in \binom{V}{t}$, and entries

$$(W_{v,k,t})_{X,T} = \begin{cases} 1, & T \subseteq X, \\ 0, & T \not\subseteq X. \end{cases}$$

Thus, for a real column vector $\mathbf{u} = (u_T)_{T \in \binom{V}{t}}$, matrix multiplication gives

$$(W_{v,k,t}\mathbf{u})_X = \sum_{T \in \binom{X}{t}} u_T.$$

The next lemma says that these sums determine $\mathbf{u}$ uniquely in the stated range.

Lemma 3.1 (Full column rank of the ordinary inclusion matrix). *If $0 \leq t \leq \min(k, v-k)$, then $W_{v,k,t}$ has full column rank over $\mathbb{R}$:*

$$\text{rank } W_{v,k,t} = \binom{v}{t}.$$

Equivalently, $W_{v,k,t}\mathbf{u} = 0$ implies $\mathbf{u} = 0$.

Proof. We prove the equivalent assertion about the kernel by induction on t , simultaneously for all admissible v and k . Suppose that

$$\sum_{T \in \binom{X}{t}} u_T = 0 \quad \text{for every } X \in \binom{V}{k}. \quad (4)$$

If $t = 0$, the only coefficient is $u_\emptyset$, and every equation in (4) gives $u_\emptyset = 0$.

Let $t \geq 1$, and assume the assertion holds with $t - 1$ in place of t . To compare coefficients indexed by t -sets differing in one element, choose distinct $x, y \in V$. For every $Z \in \binom{V \setminus \{x, y\}}{k-1}$, subtract the equations in (4) for $Z \cup \{x\}$ and $Z \cup \{y\}$. The terms indexed by t -sets contained in Z cancel, leaving

$$\sum_{D \in \binom{Z}{t-1}} (u_{D \cup \{x\}} - u_{D \cup \{y\}}) = 0.$$

These are the kernel equations for the inclusion matrix with parameters $(v-2, k-1, t-1)$. The induction hypothesis applies because

$$0 \leq t-1 \leq k-1, \quad t-1 \leq v-k-1 = (v-2) - (k-1).$$

Consequently,

$$u_{D \cup \{x\}} = u_{D \cup \{y\}} \quad \text{for every } D \in \binom{V \setminus \{x, y\}}{t-1}.$$

Since x and y were arbitrary, coefficients indexed by two t -sets differing in one element are equal. Any two t -sets can be connected by repeatedly replacing an element outside the target set with an element of the target set. Hence every coefficient u_T has one common value γ . Substituting this value into (4) gives $\binom{k}{t}\gamma = 0$. Since $\binom{k}{t} > 0$, we obtain $\gamma = 0$, as required. $\square$

To obtain the same uniqueness statement for rainbow functions, define the rainbow inclusion matrix $W_{k,t}^S$ with rows indexed by $X \in \mathcal{R}_k(S)$, columns indexed by $T \in \mathcal{R}_t(S)$, and entries

$$(W_{k,t}^S)_{X,T} = \begin{cases} 1, & T \subseteq X, \\ 0, & T \not\subseteq X. \end{cases}$$

Lemma 3.2 (Full column rank of the rainbow inclusion matrix). *Let S have m color classes, each of size $a \geq 1$. If $0 \leq t \leq \min(k, m-k)$, then $W_{k,t}^S$ has full column rank over $\mathbb{R}$:*

$$\text{rank } W_{k,t}^S = |\mathcal{R}_t(S)| = \binom{m}{t} a^t.$$

Proof. Suppose $W_{k,t}^S \mathbf{u} = 0$. We show that each coefficient u_{T_0} vanishes. Fix $T_0 \in \mathcal{R}_t(S)$, and choose one symbol from each color class, using the symbols of T_0 in the colors met by T_0 . Let V be the set of these m chosen symbols. Every subset of V is rainbow, so the equations $W_{k,t}^S \mathbf{u} = 0$ include

$$\sum_{T \in \binom{X}{t}} u_T = 0 \quad \text{for every } X \in \binom{V}{k}.$$

Lemma 3.1, applied with $v = m$, gives $u_T = 0$ for every $T \in \binom{V}{t}$. In particular, $u_{T_0} = 0$ because $T_0 \subseteq V$. Since T_0 was arbitrary, $\mathbf{u} = 0$. $\square$

The condition in Lemma 3.2 uses the number m of color classes. For the cubic parameters $t = 3$ and $q \geq 2$, we have

$$k = q + 2, \quad m - k = q^2,$$

so $3 \leq \min(k, m - k)$. Thus the ordinary and rainbow rank statements apply to all three cases in Section 2.

3.2 Functions and their unique coefficient vectors

We now turn a function into a vector by listing its values. In the ordinary setting, write $\Omega = \binom{V}{k}$, $\mathcal{T} = \binom{V}{t}$, and $W = W_{v,k,t}$; in the rainbow setting, write $\Omega = \mathcal{R}_k(S)$, $\mathcal{T} = \mathcal{R}_t(S)$, and $W = W_{k,t}^S$. Thus Ω indexes the rows and $\mathcal{T}$ indexes the columns. Let $N = |\Omega|$ and $M = |\mathcal{T}|$, and fix orderings of these two index sets. A function $f : \Omega \rightarrow \mathbb{R}$ then corresponds to the column vector

$$\mathbf{f} = (f(X))_{X \in \Omega} \in \mathbb{R}^N.$$

Each column of W is the vector of values of a star indicator b_T . Consequently, f belongs to $U_t(k)$ in the ordinary setting, or to $U_t^S(k)$ in the rainbow setting, if and only if

$$\mathbf{f} = W\mathbf{c}, \quad \mathbf{c} = (c_T)_{T \in \mathcal{T}} \in \mathbb{R}^M. \tag{5}$$

Indeed, the coordinate of $W\mathbf{c}$ indexed by X is $\sum_{T \in \binom{X}{t}} c_T$, which is exactly the star-indicator representation of f .

Under the rank conditions proved above, the coefficient vector $\mathbf{c}$ is unique. To verify this, if $W\mathbf{c} = W\mathbf{c}'$, then $W(\mathbf{c} - \mathbf{c}') = 0$. Full column rank gives $\mathbf{c} = \mathbf{c}'$.

For each column index T , let $\mathbf{e}_T \in \mathbb{R}^M$ have value 1 at coordinate T and 0 at every other coordinate. Then the vector of the star indicator b_T is

$$\mathbf{b}_T = W\mathbf{e}_T.$$

Thus, when W has full column rank, proving that $f = b_T$ is equivalent to proving that its unique coefficient vector is $\mathbf{c} = \mathbf{e}_T$.

3.3 The intersection with the Boolean hypercube

The matrix representation separates the linear restriction on f from the requirement that its values be 0 or 1. The image, or column space, of W is

$$\text{im } W = \{W\mathbf{c} : \mathbf{c} \in \mathbb{R}^M\} \subseteq \mathbb{R}^N.$$

It consists exactly of the vectors of functions in $U_t(k)$ or $U_t^S(k)$, according to the setting. The set $\{0, 1\}^N$, called the Boolean hypercube, consists of all vectors whose coordinates are 0 or 1. Hence the Boolean functions in the permitted space correspond exactly to

$$\text{im } W \cap \{0, 1\}^N.$$

For $\mathbf{z} \in \{0, 1\}^N$, its Hamming weight is the number of coordinates equal to 1:

$$\text{wt}(\mathbf{z}) = |\{X \in \Omega : z_X = 1\}| = \sum_{X \in \Omega} z_X.$$

For the vector of a Boolean function, this weight equals its support size: $\text{wt}(\mathbf{f}) = |\text{supp } f|$.

Every column $\mathbf{b}_T$ belongs to $\text{im } W \cap \{0, 1\}^N$ and has weight

$$\sigma = \begin{cases} \binom{v-t}{k-t}, & \text{in the ordinary setting,} \\ \binom{m-t}{k-t} a^{k-t}, & \text{in the balanced rainbow setting.} \end{cases}$$

Our minimum-support and classification question therefore asks whether

$$\{\mathbf{z} \in \text{im } W \cap \{0, 1\}^N : 0 < \text{wt}(\mathbf{z}) \leq \sigma\} = \{\mathbf{b}_T : T \in \mathcal{T}\}. \quad (6)$$

This equality asserts both that no nonzero Boolean vector in the image has weight below the column weight and that every Boolean vector attaining that weight is a column. When the columns are linearly independent, the corresponding question in coefficient space is whether $W\mathbf{c} \in \{0, 1\}^N$ and $0 < \text{wt}(W\mathbf{c}) \leq \sigma$ force $\mathbf{c} = \mathbf{e}_T$ for some T . The counterexample at $q = 2$ gives an additional point on the left-hand side of (6); our positive results at $q = 3$ and at $q = 4$ prove the displayed equality for their respective parameters.

4 A Counterexample for $t = 3$ and $q = 2$

Proof of Theorem 2.1. We construct the required function on $\Omega = \binom{[8]}{4}$. For $X \in \Omega$, let $x_i(X) = 1$ if $i \in X$ and $x_i(X) = 0$ otherwise. Put $r(X) = x_1(X) + x_2(X) + x_3(X) +$

$x_4(X) = |X \cap [4]|$, and define

$$f(X) = x_8(X) \frac{(r(X) - 1)(r(X) - 2)}{2}. \quad (7)$$

To verify that f is Boolean and determine its support, first consider X with $8 \notin X$. Then $x_8(X) = 0$, so $f(X) = 0$. If $8 \in X$, the other three elements of X give $r(X) \in \{0, 1, 2, 3\}$, and

$$\frac{(r(X) - 1)(r(X) - 2)}{2} = \begin{cases} 1, & r(X) \in \{0, 3\}, \\ 0, & r(X) \in \{1, 2\}. \end{cases}$$

Thus f takes only the values 0 and 1. When $8 \in X$ and $r(X) = 0$, the other three elements must be 5, 6, 7. When $8 \in X$ and $r(X) = 3$, the other three elements form a triple in $[4]$. Hence

$$\mathcal{F} := \text{supp } f = \{\{1, 2, 3, 8\}, \{1, 2, 4, 8\}, \{1, 3, 4, 8\}, \{2, 3, 4, 8\}, \{5, 6, 7, 8\}\}.$$

These five sets are distinct, so

$$|\text{supp } f| = 5 = \binom{8-3}{4-3},$$

which is the size of a 3-star on Ω .

To check the degree bound, observe that $r(X)$ is linear in the membership variables $x_i(X)$. Formula (7) is therefore a polynomial of total degree at most three: a quadratic polynomial in $r(X)$ multiplied by $x_8(X)$. By the polynomial characterization in Section 1, $f \in U_3(4)$. Its representation using only 3-star indicators follows from identity (3), and the triple coefficient vector is unique by Lemma 3.1, since $3 \leq \min(4, 8 - 4)$.

To show that f is not a 3-star indicator, we examine the elements common to all sets in $\mathcal{F}$. Observe that

$$\bigcap_{X \in \mathcal{F}} X = \{8\}.$$

If $f = b_T$ for a triple T , then every member of $\mathcal{F}$ would contain T . This would force $T \subseteq \{8\}$, contradicting $|T| = 3$. Thus f has all the properties asserted in Theorem 2.1. $\square$

5 Star Conclusion for $t = 3$, $q = 3$, and $a \geq 1$

We prove Theorem 2.2 in three stages. We first work on the ordinary slice $\binom{[14]}{5}$. In Section 5.1, we prove the support bound 55 and show that equality forces the function to vanish on every five-set meeting a support set in 0 or 2 elements. These zero values give linear equations on the triple coefficients. In Section 5.2, we use those equations to show that every function attaining support 55 has exactly one coefficient equal to 1 and

all the others equal to 0, giving an ordinary 3-star indicator. Finally, in Section 5.3, we extend the support bound and star classification to 14 color classes of arbitrary common size $a \geq 1$.

5.1 The ordinary support bound and its equality consequences

Proposition 5.1 (Support bound and forced zero values). *Let $\Omega = \binom{[14]}{5}$. Every nonzero Boolean function $f \in U_3(5)$ on Ω satisfies*

$$|\text{supp } f| \geq 55.$$

If $|\text{supp } f| = 55$, then for every $A \in \text{supp } f$,

$$f(Y) = 0 \quad \text{whenever } Y \in \Omega \text{ and } |A \cap Y| \in \{0, 2\}. \quad (8)$$

Proof. Every triple $T \in \binom{[14]}{3}$ is contained in exactly $\binom{11}{2} = 55$ members of Ω : choose the other two elements outside T . Using the notation of Section 3, let $W = W_{14,5,3}$ and write

$$\mathbf{f} = W\mathbf{c} = \sum_{T \in \binom{[14]}{3}} c_T \mathbf{b}_T,$$

where $\mathbf{f}$ is the vector of values of f and $\mathbf{b}_T$ is the column of W corresponding to the star indicator b_T . Summing the coordinates and using the count above gives

$$\begin{aligned} |\text{supp } f| &= \sum_{X \in \Omega} f(X) = \sum_{X \in \Omega} \sum_{T \in \binom{X}{3}} c_T \\ &= 55 \sum_{T \in \binom{[14]}{3}} c_T. \end{aligned}$$

Our first goal is therefore to prove

$$\sum_{T \in \binom{[14]}{3}} c_T \geq 1.$$

The coefficients c_T may be negative. To bound their sum using the nonnegative values of f , we seek a matrix H , with rows and columns indexed by Ω , whose entries are nonnegative and which satisfies

$$H\mathbf{f} = \left(\sum_{T \in \binom{[14]}{3}} c_T \right) \mathbf{1} - \mathbf{f}, \quad (9)$$

where $\mathbf{1}$ is the vector with value 1 at every coordinate indexed by Ω . Such a matrix would prove the desired inequality. Indeed, choose $A \in \text{supp } f$, so that $f(A) = 1$. The coordinate of (9) indexed by A would give

$$0 \leq \sum_{Y \in \Omega} H_{A,Y} f(Y) = (H\mathbf{f})_A = \sum_{T \in \binom{[14]}{3}} c_T - 1.$$

The first inequality follows from $H_{A,Y} \geq 0$ and $f(Y) \geq 0$. Thus constructing H would give $\sum_T c_T \geq 1$, and hence $|\text{supp } f| \geq 55$.

To make one matrix H work for every function in $U_3(5)$, it suffices to require

$$H\mathbf{b}_T = \mathbf{1} - \mathbf{b}_T \quad \text{for every } T \in \binom{[14]}{3}. \quad (10)$$

Indeed, linearity would then give

$$H\mathbf{f} = \sum_T c_T H\mathbf{b}_T = \sum_T c_T (\mathbf{1} - \mathbf{b}_T) = \left(\sum_T c_T \right) \mathbf{1} - \mathbf{f},$$

which is (9). We now construct H by imposing the identities in (10).

First, nonnegativity and (10) force $H_{X,Y} = 0$ whenever $|X \cap Y| \geq 3$. To see this, choose a triple $T \subseteq X \cap Y$. Since $b_T(X) = 1$, the coordinate indexed by X must satisfy

$$0 = (H\mathbf{b}_T)_X = \sum_{\substack{Y' \in \Omega \\ T \subseteq Y'}} H_{X,Y'}.$$

Every summand is nonnegative, and $H_{X,Y}$ is one of these summands because $T \subseteq Y$. Therefore $H_{X,Y} = 0$.

For the remaining intersection sizes, we look for weights that are unchanged by relabeling the elements of $[14]$. We therefore assign a common weight to each intersection size and seek $\alpha, \beta, \gamma \geq 0$ such that

$$H_{X,Y} = \begin{cases} \alpha, & |X \cap Y| = 2, \\ \beta, & |X \cap Y| = 1, \\ \gamma, & |X \cap Y| = 0, \\ 0, & |X \cap Y| \geq 3. \end{cases}$$

We determine these three weights by counting the sets that contribute to each coordinate of $H\mathbf{b}_T$.

Fix $X \in \Omega$ and $T \in \binom{[14]}{3}$, and put $r = |T \cap X|$. Then

$$(H\mathbf{b}_T)_X = \sum_{\substack{Y \in \Omega \\ T \subseteq Y}} H_{X,Y}.$$

If $r = 3$, every Y containing T meets X in at least three elements, so the sum is 0, as required by $1 - b_T(X) = 0$. For $r = 0, 1, 2$, we have $b_T(X) = 0$, so the required sum is 1.

Suppose first that $r = 2$. Only sets Y with $|X \cap Y| = 2$ can contribute: smaller intersections are impossible because $T \subseteq Y$, and larger intersections have weight 0. To form such a Y , add two elements to T , both outside $X \cup T$. The union $X \cup T$ has six elements, leaving eight elements from which to choose. Hence

$$\binom{8}{2}\alpha = 28\alpha = 1, \quad \text{so } \alpha = \frac{1}{28}.$$

Suppose next that $r = 1$. To obtain $|X \cap Y| = 2$, add one element from $X \setminus T$ and one from outside $X \cup T$. There are four choices for the first and seven for the second, giving 28 sets. To obtain $|X \cap Y| = 1$, add both elements outside $X \cup T$, giving $\binom{7}{2} = 21$ sets. Intersection size 0 is impossible. Thus

$$28\alpha + 21\beta = 1.$$

Since $28\alpha = 1$, this forces $\beta = 0$.

Finally, suppose that $r = 0$. The sets X and T are disjoint, and six elements lie outside $X \cup T$. Adding two elements of X gives $\binom{5}{2} = 10$ sets with intersection size 2. Adding one element of X and one outside $X \cup T$ gives $5 \cdot 6 = 30$ sets with intersection size 1. Adding both elements outside $X \cup T$ gives $\binom{6}{2} = 15$ sets with intersection size 0. The required equation is therefore

$$10\alpha + 30\beta + 15\gamma = 1.$$

Substituting $\alpha = 1/28$ and $\beta = 0$ gives $\gamma = 3/70$.

We have obtained the matrix

$$H_{X,Y} = \begin{cases} \frac{1}{28}, & |X \cap Y| = 2, \\ \frac{3}{70}, & |X \cap Y| = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

Its entries are nonnegative, and the four cases for r verify (10) for every coordinate X and every triple T . Thus (9) holds, and the argument given above proves $|\text{supp } f| \geq 55$.

It remains to derive the zero values forced by equality. Assume $|\text{supp } f| = 55$. The

initial counting identity gives $\sum_T c_T = 1$, so (9) becomes $H\mathbf{f} = \mathbf{1} - \mathbf{f}$. For any $A \in \text{supp } f$, its coordinate therefore gives

$$0 = (H\mathbf{f})_A = \frac{1}{28} \sum_{\substack{Y \in \Omega \\ |A \cap Y| = 2}} f(Y) + \frac{3}{70} \sum_{\substack{Y \in \Omega \\ |A \cap Y| = 0}} f(Y).$$

Every term is nonnegative, and both weights are strictly positive. Hence $f(Y) = 0$ whenever $|A \cap Y|$ is 0 or 2, proving (8). $\square$

5.2 Classification of the ordinary equality case

Proposition 5.1 reduces the ordinary classification to Boolean functions with support size 55. We will examine five-sets that meet a support set in exactly one element. After fixing that element, the other four elements are chosen from the nine-element complement of the support set. The following lemma gives a support lower bound for nonzero Boolean functions of degree at most one on $\binom{R}{4}$, where $|R| = 9$.

Lemma 5.2 (Boolean functions of degree at most one on a smaller slice). *Let R be a nine-element set, and let*

$$h(Z) = \beta + \sum_{u \in Z} \alpha_u \quad \left(Z \in \binom{R}{4} \right)$$

be a nonzero Boolean function, where β and the α_u are real numbers. Then

$$|\text{supp } h| \geq 56.$$

Proof. We first compare coefficients by exchanging one element of an input. For distinct $u, v \in R$, choose $B \in \binom{R \setminus \{u, v\}}{3}$. Then

$$\alpha_u - \alpha_v = h(B \cup \{u\}) - h(B \cup \{v\}) \in \{-1, 0, 1\},$$

because both function values belong to $\{0, 1\}$. If all coefficients α_u are equal, the sum on every four-element set is the same. Since h is nonzero and Boolean, it must be the constant function 1.

Otherwise, put $\lambda = \min_{u \in R} \alpha_u$. Comparing any coefficient with one attaining this minimum shows that every coefficient is either λ or $\lambda + 1$. Let

$$P = \{u \in R : \alpha_u = \lambda + 1\}.$$

The set P is nonempty and is not all of R . Since each input has four elements, the

common contribution 4λ can be absorbed into the constant term. Thus

$$h(Z) = \gamma + |Z \cap P|, \quad \gamma = \beta + 4\lambda.$$

We now use Booleanity to determine the possible sizes of P .

Since h takes only the values 0 and 1, the intersection size $|Z \cap P|$ can take at most two values. This forces $|P| = 1$ or $|P| = 8$: otherwise, four-element sets meet P in at least three different numbers of elements.

If $|P| = 1$, write $P = \{u_0\}$. The intersection sizes 0 and 1 both occur, so γ and $\gamma + 1$ must be 0 and 1, respectively. Thus $h(Z) = 1$ exactly when $u_0 \in Z$. If $|P| = 8$, write $R \setminus P = \{u_0\}$. The intersection sizes are 3 and 4, so $\gamma + 3 = 0$ and $\gamma + 4 = 1$. Consequently, $h(Z) = 1$ exactly when $u_0 \notin Z$.

The nonzero constant function has support $\binom{9}{4} = 126$. A membership indicator has support $\binom{8}{3} = 56$, because the other three elements are chosen from the remaining eight. A nonmembership indicator has support $\binom{8}{4} = 70$. The smallest of these three support sizes is 56. $\square$

Theorem 5.3 (Classification on the ordinary slice). *Let $f \in U_3(5)$ on $\binom{[14]}{5}$ be Boolean. If $0 < |\text{supp } f| \leq 55$, then $f = b_{T_0}$ for a unique triple $T_0 \in \binom{[14]}{3}$.*

Proof. Proposition 5.1 gives $|\text{supp } f| = 55$. Fix $A \in \text{supp } f$ and put $R = [14] \setminus A$, so $|A| = 5$ and $|R| = 9$. As in Section 3, write $\mathbf{f} = W_{14,5,3}\mathbf{c}$ using the unique triple coefficient vector $\mathbf{c}$. In coordinates,

$$f(X) = \sum_{T \in \binom{X}{3}} c_T \quad \left(X \in \binom{[14]}{5} \right). \quad (12)$$

Each triple is contained in 55 five-sets, so summing this representation gives

$$\sum_{T \in \binom{[14]}{3}} c_T = \frac{|\text{supp } f|}{55} = 1. \quad (13)$$

We will use the zero values in (8) to determine this coefficient vector.

Step 1: use the zero values on sets disjoint from A . For every $X \in \binom{R}{5}$, Proposition 5.1 gives

$$0 = f(X) = \sum_{T \in \binom{X}{3}} c_T.$$

These equations say that the vector of coefficients indexed by triples in R belongs to the kernel of $W_{9,5,3}$. Lemma 3.1 applies because $3 \leq \min(5, 9 - 5)$, and therefore

$$c_T = 0 \quad \left(T \in \binom{R}{3} \right). \quad (14)$$

Step 2: use the zero values on sets meeting A in two elements. To describe the remaining triples that meet both A and R , define

$$a_i(\{u, v\}) = c_{\{i, u, v\}}, \quad b_{ij}(u) = c_{\{i, j, u\}},$$

where $i, j \in A$ are distinct and $u, v \in R$ are distinct. Thus a_i is a function on unordered pairs in R , while b_{ij} is a function on elements of R ; also $b_{ij} = b_{ji}$. For distinct $i, j \in A$ and every $Z \in \binom{R}{3}$, the set $\{i, j\} \cup Z$ meets A in two elements, so its value under f is zero. The triple contained entirely in Z has coefficient zero by (14). Hence

$$0 = \sum_{\{u, v\} \in \binom{Z}{2}} (a_i(\{u, v\}) + a_j(\{u, v\})) + \sum_{u \in Z} b_{ij}(u). \quad (15)$$

To apply the rank lemma, we express the entire right-hand side as a sum indexed by pairs. Each element of the three-element set Z belongs to two of its pairs, so

$$\sum_{u \in Z} b_{ij}(u) = \frac{1}{2} \sum_{\{u, v\} \in \binom{Z}{2}} (b_{ij}(u) + b_{ij}(v)).$$

Substituting into (15) gives

$$\sum_{\{u, v\} \in \binom{Z}{2}} \left[a_i(\{u, v\}) + a_j(\{u, v\}) + \frac{1}{2}(b_{ij}(u) + b_{ij}(v)) \right] = 0 \quad \left(Z \in \binom{R}{3} \right).$$

For fixed i, j , these are kernel equations for $W_{9,3,2}$, with the bracketed quantities as pair coefficients. Since $2 \leq \min(3, 9-3)$, Lemma 3.1 forces each bracketed quantity to vanish. Thus, for all distinct $i, j \in A$ and distinct $u, v \in R$,

$$a_i(\{u, v\}) + a_j(\{u, v\}) = -\frac{1}{2}(b_{ij}(u) + b_{ij}(v)). \quad (16)$$

Step 3: apply the Boolean linear-function lemma to sets meeting A in one element. Fix $i \in A$, and define

$$g_i(Z) = f(\{i\} \cup Z) \quad \left(Z \in \binom{R}{4} \right).$$

The function g_i is Boolean. The map $Z \mapsto \{i\} \cup Z$ sends distinct inputs to distinct five-sets, so

$$|\text{supp } g_i| \leq |\text{supp } f| = 55.$$

By (14), triples contained entirely in Z contribute nothing. Therefore

$$g_i(Z) = \sum_{\{u,v\} \in \binom{Z}{2}} a_i(\{u,v\}).$$

To apply Lemma 5.2, we want to express these pair coefficients as sums of terms depending on single elements. That is, we seek a function $p_i : R \rightarrow \mathbb{R}$ such that $a_i(\{u,v\}) = p_i(u) + p_i(v)$. Equation (16) provides such decompositions for sums $a_i + a_j$. We can isolate a_i by combining three of those sums. For this fixed i , choose any distinct $j, \ell \in A \setminus \{i\}$ and keep them fixed as u, v vary. Then

$$\begin{aligned} 2a_i(\{u,v\}) &= (a_i(\{u,v\}) + a_j(\{u,v\})) + (a_i(\{u,v\}) + a_\ell(\{u,v\})) \\ &\quad - (a_j(\{u,v\}) + a_\ell(\{u,v\})). \end{aligned}$$

Substituting (16) into these three terms gives the desired decomposition with

$$p_i(u) = -\frac{1}{4}(b_{ij}(u) + b_{i\ell}(u) - b_{j\ell}(u)).$$

We suppress these fixed choices in the notation p_i ; for each i , one arbitrary choice of j, ℓ is sufficient. In particular,

$$a_i(\{u,v\}) = p_i(u) + p_i(v) \quad \left(\{u,v\} \in \binom{R}{2} \right). \quad (17)$$

Consequently,

$$g_i(Z) = \sum_{\{u,v\} \in \binom{Z}{2}} (p_i(u) + p_i(v)) = 3 \sum_{u \in Z} p_i(u). \quad (18)$$

Indeed, each element of a four-element set belongs to three of its pairs. Thus g_i has degree at most one. If g_i were nonzero, Lemma 5.2 would give $|\text{supp } g_i| \geq 56$, contradicting its support bound 55. Hence g_i is identically zero for every $i \in A$.

Step 4: eliminate the remaining coefficients outside A and recover the star. Our remaining goal is to prove

$$c_T = 0 \quad \text{whenever } T \not\subseteq A. \quad (19)$$

This would finish the classification. Indeed, if (19) holds, take $T \in \binom{A}{3}$ and choose $Z \in \binom{R}{2}$. The set $T \cup Z$ contains exactly one triple lying entirely in A , namely T . Thus its coefficient sum reduces to

$$c_T = f(T \cup Z) \in \{0, 1\}.$$

Together with (13), this would force exactly one triple coefficient to equal 1 and all the others to equal 0. It therefore suffices to establish (19).

For each $i \in A$, the conclusion $g_i = 0$ and (18) give

$$\sum_{u \in Z} p_i(u) = 0 \quad \left(Z \in \binom{R}{4} \right).$$

These are the kernel equations for $W_{9,4,1}$. Lemma 3.1 applies because $1 \leq \min(4, 9 - 4)$, so $p_i(u) = 0$ for every $u \in R$. Equation (17) now gives $a_i(\{u, v\}) = 0$ for every pair in R .

Equation (16) then gives

$$b_{ij}(u) + b_{ij}(v) = 0 \quad \left(\{u, v\} \in \binom{R}{2} \right)$$

for each pair of distinct $i, j \in A$. These are the kernel equations for $W_{9,2,1}$. Since $1 \leq \min(2, 9 - 2)$, another application of Lemma 3.1 gives $b_{ij}(u) = 0$ for every $u \in R$.

Step 1 gives $c_T = 0$ for triples entirely in R . For triples with exactly one element in A , their coefficients are $c_{\{i,u,v\}} = a_i(\{u, v\}) = 0$; for triples with exactly two elements in A , their coefficients are $c_{\{i,j,u\}} = b_{ij}(u) = 0$. These are all the triples $T \not\subseteq A$, so (19) holds. The coefficient argument above therefore gives $\mathbf{c} = \mathbf{e}_{T_0}$ for a triple $T_0 \subseteq A$, and hence

$$\mathbf{f} = W_{14,5,3} \mathbf{e}_{T_0} = \mathbf{b}_{T_0}.$$

Thus $f = b_{T_0}$. The uniqueness of the triple coefficient vector also makes T_0 unique. $\square$

5.3 Extension to balanced rainbow slices

To apply the ordinary results to a rainbow function, we will choose one symbol from each color class and restrict the function to five-element subsets of the chosen symbols. Each restriction is an ordinary function on a fourteen-element ground set. To control how many of these restrictions are nonzero, we first prove a support bound for functions of independent choices.

For a positive integer a and a nonnegative integer m , let $[a]^m$ be the set of tuples $z = (z_1, \dots, z_m)$ with $z_i \in [a]$. We put $[0] = \emptyset$ and regard $[a]^0$ as the set containing one empty tuple. A function on $[a]^m$ depends only on coordinates in $I \subseteq [m]$ if its value is unchanged whenever the coordinates outside I are changed. For any real function h on this domain, write $\text{supp } h = \{z \in [a]^m : h(z) \neq 0\}$.

Lemma 5.4 (Support bound on a finite product). *Let $a \geq 1$ and $0 \leq t \leq m$ be integers. Suppose that a nonzero real function h on $[a]^m$ is a finite sum of functions, each depending*

on at most t coordinates. Then

$$|\text{supp } h| \geq a^{m-t}.$$

Proof. If $a = 1$, the domain has one element and the assertion is immediate. Assume $a \geq 2$. We proceed by induction on t , simultaneously for all $m \geq t$. If $t = 0$, every summand is constant, so h is a nonzero constant and has support size a^m .

Let $t \geq 1$, and assume the assertion holds with $t - 1$ in place of t . If h is constant, its support size is $a^m \geq a^{m-t}$. Otherwise, some change in a single coordinate changes the value of h . Indeed, two tuples with different values of h can be joined by changing one coordinate at a time. After relabeling the coordinates, we may assume that the last coordinate has this property. Thus there are distinct $b, b' \in [a]$ for which

$$\Delta(y) = h(y, b) - h(y, b') \quad (y \in [a]^{m-1})$$

is not identically zero.

This difference reduces the number of coordinates on which each summand can depend. A summand independent of the last coordinate cancels; a summand depending on that coordinate contributes a function of at most $t - 1$ of the remaining coordinates. Hence the induction hypothesis applies to Δ and gives

$$|\text{supp } \Delta| \geq a^{(m-1)-(t-1)} = a^{m-t}.$$

For each $y \in \text{supp } \Delta$, at least one of $h(y, b)$ and $h(y, b')$ is nonzero. Distinct choices of y give distinct tuples in $\text{supp } h$. Therefore $|\text{supp } h| \geq |\text{supp } \Delta| \geq a^{m-t}$. $\square$

Proof of Theorem 2.2. Let S have color classes $C_1, \dots, C_{14}$, each of size a , and let $f \in U_3^S(5)$ be nonzero and Boolean. Label the symbols in C_i by $(i, 1), \dots, (i, a)$. Write the triple coefficient representation of f as

$$f(X) = \sum_{T \in \binom{[14]}{3}} c_T \quad (X \in \mathcal{R}_5(S)), \quad (20)$$

where the real coefficients c_T are indexed by $T \in \mathcal{R}_3(S)$.

We first construct the ordinary restrictions to which Proposition 5.1 and Theorem 5.3 apply. For each $z \in [a]^{14}$, choose the set

$$V_z = \{(1, z_1), \dots, (14, z_{14})\},$$

and define

$$f_z(X) = f(X) \quad \left(X \in \binom{V_z}{5} \right).$$

Every subset of V_z is rainbow. Consequently, f_z is Boolean, and (20) represents f_z using the coefficients indexed by triples contained in V_z . Thus $f_z \in U_3(5)$ on the ordinary slice $\binom{V_z}{5}$.

To record the support sizes of all these restrictions in one function, define

$$h(z) = \frac{|\text{supp } f_z|}{55} = \frac{1}{55} \sum_{X \in \binom{V_z}{5}} f(X) = \sum_{T \in \binom{V_z}{3}} c_T \quad (z \in [a]^{14}). \quad (21)$$

The middle equality uses Booleanity. To verify the last equality, substitute (20) and note that each triple in V_z is contained in exactly $\binom{11}{2} = 55$ five-element subsets of V_z .

The coefficient expression for h allows us to apply Lemma 5.4. More explicitly,

$$h(z) = \sum_{1 \leq i < j < \ell \leq 14} c_{\{(i, z_i), (j, z_j), (\ell, z_\ell)\}}.$$

For fixed i, j, ℓ , the corresponding summand depends only on z_i, z_j, z_ℓ . Thus h is a sum of functions depending on at most three coordinates. Moreover, h is nonzero: choose $X \in \text{supp } f$ and extend its five symbols to a set V_z by choosing one symbol from each of the other nine color classes. Then $f_z(X) = 1$, so $h(z) > 0$. Lemma 5.4, with $m = 14$ and $t = 3$, now gives $|\text{supp } h| \geq a^{11}$.

We obtain the desired lower bound for f from the following chain:

$$a^{11} \leq |\text{supp } h| \leq \sum_{z \in [a]^{14}} h(z) = \frac{a^9}{55} |\text{supp } f|. \quad (22)$$

The first inequality was just proved. For the second, Proposition 5.1 says that each restriction f_z is either zero or has support size at least 55. By (21), this means $h(z) = 0$ or $h(z) \geq 1$. Hence every tuple in $\text{supp } h$ contributes at least 1 to the sum of the values of h .

To verify the final equality in (22), count pairs (z, X) with $X \in \text{supp } f$ and $X \subseteq V_z$. For fixed z , the number of such sets X is $|\text{supp } f_z|$. For fixed X , the choices of z in the five colors used by X are prescribed, while the other nine coordinates are arbitrary. Thus each X belongs to exactly a^9 of the sets V_z , and

$$\sum_{z \in [a]^{14}} h(z) = \frac{1}{55} \sum_{z \in [a]^{14}} |\text{supp } f_z| = \frac{a^9}{55} |\text{supp } f|.$$

Dividing (22) by $a^9/55$ proves $|\text{supp } f| \geq 55a^2$.

Now suppose that $|\text{supp } f| = 55a^2$. We want to show that exactly one triple coefficient equals 1 and all the others equal 0. First, summing (20) over the entire rainbow slice

gives

$$|\text{supp } f| = 55a^2 \sum_{T \in \mathcal{R}_3(S)} c_T, \quad \text{and hence} \quad \sum_{T \in \mathcal{R}_3(S)} c_T = 1. \quad (23)$$

Indeed, each rainbow triple extends to a rainbow five-element set by choosing two of the remaining eleven colors and one symbol from each, giving $\binom{11}{2}a^2 = 55a^2$ extensions. In view of (23), it therefore suffices to prove $c_T \in \{0, 1\}$ for every rainbow triple T . We will obtain these individual coefficient values from the ordinary restrictions.

Under the equality assumption, both ends of (22) equal a^{11} , so every inequality in that chain is an equality. In particular,

$$0 = \sum_{z \in [a]^{14}} h(z) - |\text{supp } h| = \sum_{z \in \text{supp } h} (h(z) - 1).$$

Every summand is nonnegative by the argument for the second inequality. Hence $h(z) = 1$ whenever $h(z) \neq 0$, and h is Boolean. It follows from (21) that each f_z is either zero or has support size 55. Theorem 5.3 identifies every nonzero f_z as an ordinary 3-star indicator.

To translate this classification into coefficient values, fix z and let $\mathbf{c}_z = (c_T)_{T \in \binom{V_z}{3}}$. Using the vector notation of Section 3, the restriction has value vector

$$\mathbf{f}_z = W_{14,5,3} \mathbf{c}_z,$$

where the rows and columns of this inclusion matrix are indexed by five-element subsets and triples of V_z . By Lemma 3.1, the coefficient vector $\mathbf{c}_z$ is unique. If $f_z = 0$, this vector is zero. If $f_z = b_{T_z}$ for a triple $T_z \subseteq V_z$, this vector is the standard basis vector $\mathbf{e}_{T_z}$, whose only nonzero entry is 1 at T_z . Therefore $c_T \in \{0, 1\}$ for every triple $T \subseteq V_z$.

Every rainbow triple T is contained in some V_z : choose its symbols in its three colors and any symbols in the remaining eleven colors. The preceding conclusion therefore gives $c_T \in \{0, 1\}$ for every $T \in \mathcal{R}_3(S)$. Together with (23), this forces a single coefficient $c_{T_0} = 1$ and all other coefficients to be zero. Substituting into (20) yields $f = b_{T_0}$ on $\mathcal{R}_5(S)$.

Conversely, every rainbow 3-star indicator belongs to $U_3^S(5)$, is Boolean, and has support size $55a^2$ by (1). This proves both the support bound and the equality classification. $\square$

6 Star Conclusion for $t = 3$, $q = 4$, and $a \geq 1$

For $q = 4$, the ordinary parameters are $(m, k, t) = (22, 6, 3)$, and a triple star has size $\binom{19}{3} = 969$. We follow the same three stages as in Section 5. First, we prove the ordinary support bound and the zero values forced by equality. Next, we determine the ordinary

equality case by examining coefficients relative to a fixed support set. Finally, we apply the restriction argument from Subsection 5.3 to prove Theorem 2.3 for every $a \geq 1$.

6.1 The ordinary support bound and equality constraints

Proposition 6.1 (The ordinary support bound and forced zero values). *Let $\Omega = \binom{[22]}{6}$, and let $f \in U_3(6)$ on Ω be a nonzero Boolean function. Then*

$$|\text{supp } f| \geq 969.$$

If $|\text{supp } f| = 969$, then for every $A \in \text{supp } f$,

$$f(Y) = 0 \quad \text{whenever } Y \in \Omega \text{ and } |A \cap Y| \in \{0, 2\}. \quad (24)$$

We begin by explaining the counting tool needed for this proposition. If $f(X) = \sum_{T \in \binom{X}{3}} c_T$, then each triple belongs to 969 six-element sets, so $|\text{supp } f| = 969 \sum_T c_T$. To prove the lower bound, it therefore suffices to show $\sum_T c_T \geq 1$. We will express this coefficient sum as a sum of Boolean values over a collection of six-element sets. For the coefficient of every c_T in that sum to be 1, each triple must occur in exactly one set of the collection.

Such a collection is provided by the classical twenty-two-point Witt design [2, Theorem 7.3]. More precisely, there exists a collection $\mathcal{D} \subseteq \binom{[22]}{6}$ such that every triple of $[22]$ is contained in exactly one member of $\mathcal{D}$. A collection with this property is called a Steiner system $S(3, 6, 22)$, and its members are called blocks. Fix one such design $\mathcal{D}$. We will use its triple-covering property to prove the support bound, and the following intersection counts to derive the zero values forced by equality.

Lemma 6.2 (Counting properties of the Witt design). *The design $\mathcal{D}$ has 77 blocks. For every block $A \in \mathcal{D}$, exactly 16 other blocks are disjoint from A , exactly 60 other blocks meet A in two points, and no other intersection size occurs between distinct blocks.*

Proof. Every triple lies in exactly one block, and each block contains $\binom{6}{3} = 20$ triples. Counting pairs consisting of a triple and a block containing it gives

$$|\mathcal{D}| = \frac{\binom{22}{3}}{\binom{6}{3}} = \frac{1540}{20} = 77.$$

We also need the number of blocks through a fixed point or pair. There are $\binom{21}{2} = 210$ triples through a fixed point, and each block through that point contains $\binom{5}{2} = 10$ such triples. Hence the point lies in 21 blocks. There are 20 triples through a fixed pair, and each block through that pair contains 4 such triples. Hence the pair lies in 5 blocks.

Distinct blocks cannot share three points, since the triple formed by those points would occur in two blocks. Fix $A \in \mathcal{D}$, and let n_r count the other blocks meeting A in r points, for $r = 0, 1, 2$. There are 76 other blocks, so

$$n_0 + n_1 + n_2 = 76.$$

To count their incidences with points of A , note that each of the six points of A lies in 20 blocks other than A . A block meeting A in r points contributes r incidences. Therefore

$$n_1 + 2n_2 = 6 \cdot 20 = 120.$$

Finally, each of the $\binom{6}{2} = 15$ pairs in A lies in 4 blocks other than A . A block meeting A in two points contributes exactly one such pair, and a block meeting A in fewer points contributes none. Thus

$$n_2 = \binom{6}{2} \cdot 4 = 60.$$

The last two equations give $n_1 = 0$, and the first then gives $n_0 = 16$. □

Proof of Proposition 6.1. Using Section 3, write $\mathbf{f} = W_{22,6,3}\mathbf{c}$. In coordinates,

$$f(X) = \sum_{T \in \binom{X}{3}} c_T \quad (X \in \Omega).$$

Summing over all six-element sets gives

$$|\text{supp } f| = 969 \sum_{T \in \binom{[22]}{3}} c_T. \tag{25}$$

Let $\mathcal{D}$ be the Witt design fixed above. Since each triple belongs to exactly one member of $\mathcal{D}$,

$$\sum_{D \in \mathcal{D}} f(D) = \sum_{T \in \binom{[22]}{3}} c_T = \frac{|\text{supp } f|}{969}. \tag{26}$$

The left-hand side is an integer because f is Boolean, and the right-hand side is strictly positive because f is nonzero. Hence $|\text{supp } f|/969 \geq 1$, proving the support bound.

Suppose now that $|\text{supp } f| = 969$. For any permutation π of $[22]$, the collection $\pi(\mathcal{D}) = \{\pi(D) : D \in \mathcal{D}\}$ still covers every triple exactly once. Applying (26) to this collection gives

$$\sum_{D \in \pi(\mathcal{D})} f(D) = 1.$$

Thus each relabeled collection contains exactly one member of $\text{supp } f$. We will use this fact to rule out intersections of size 0 or 2 between two support sets.

By Lemma 6.2, there are distinct $D_1, D_2 \in \mathcal{D}$ meeting in 0 elements, and there are distinct members meeting in 2 elements. Suppose that distinct $A, Y \in \text{supp } f$ satisfy $|A \cap Y| = r \in \{0, 2\}$. Choose $D_1, D_2 \in \mathcal{D}$ with $|D_1 \cap D_2| = r$. A permutation π can send the ordered pair (D_1, D_2) to (A, Y) . Indeed, the four disjoint parts

$$D_1 \cap D_2, \quad D_1 \setminus D_2, \quad D_2 \setminus D_1, \quad [22] \setminus (D_1 \cup D_2)$$

have sizes $r, 6 - r, 6 - r, 10 + r$, respectively, as do the corresponding parts formed from A, Y . Choose a bijection between each pair of corresponding parts and combine the four bijections. Then A and Y are two support sets in $\pi(\mathcal{D})$, contradicting the sum 1 above. Therefore no such pair of support sets exists. Since f is Boolean, this proves the zero values in (24). $\square$

6.2 Classification of the ordinary equality case

Proposition 6.1 reduces the ordinary classification to Boolean functions with support size 969. As in Subsection 5.2, we will fix a support set and examine inputs meeting it in exactly one element. The other five elements then lie in the sixteen-element complement of the support set. The following lemma gives the support bound needed for a linear function on this smaller slice. Unlike Lemma 5.2, this bound does not require Booleanity.

Lemma 6.3 (Support bound for a linear function on a smaller slice). *Let R be a sixteen-element set, and let*

$$g(Z) = \beta + \sum_{u \in Z} \alpha_u \quad \left(Z \in \binom{R}{5} \right)$$

be a nonzero real function, where β and the α_u are real numbers. Then

$$|\text{supp } g| \geq \binom{14}{4} = 1001.$$

Proof. If all coefficients α_u are equal, then g is constant. Since g is nonzero, every input belongs to its support, giving $|\text{supp } g| = \binom{16}{5} = 4368 \geq 1001$.

Otherwise, choose distinct $u, v \in R$ with $\alpha_u \neq \alpha_v$. For each $B \in \binom{R \setminus \{u, v\}}{4}$, compare the two inputs $B \cup \{u\}$ and $B \cup \{v\}$:

$$g(B \cup \{u\}) - g(B \cup \{v\}) = \alpha_u - \alpha_v \neq 0.$$

Their values cannot both be zero, so at least one of these inputs belongs to $\text{supp } g$. Different choices of B give disjoint pairs of inputs: the first input contains u but not v , the second contains v but not u , and deleting the distinguished element recovers B . Thus there is at least one distinct support point for each of the $\binom{14}{4} = 1001$ choices of B . $\square$

Theorem 6.4 (Classification on the ordinary slice). *Let $f \in U_3(6)$ on $\binom{[22]}{6}$ be Boolean. If $0 < |\text{supp } f| \leq 969$, then $f = b_{T_0}$ for a unique triple $T_0 \in \binom{[22]}{3}$.*

Proof. Proposition 6.1 gives $|\text{supp } f| = 969$. Fix $A \in \text{supp } f$ and put $R = [22] \setminus A$, so $|A| = 6$ and $|R| = 16$. Write $\mathbf{f} = W_{22,6,3}\mathbf{c}$ using the unique triple coefficient vector. In coordinates,

$$f(X) = \sum_{T \in \binom{X}{3}} c_T \quad \left(X \in \binom{[22]}{6} \right). \quad (27)$$

Each triple is contained in $\binom{[19]}{3} = 969$ six-element sets. Summing the representation over all inputs therefore gives

$$\sum_{T \in \binom{[22]}{3}} c_T = \frac{|\text{supp } f|}{969} = 1. \quad (28)$$

We will use the zero values in (24) to determine this coefficient vector. The four steps follow the argument in Subsection 5.2; we include the equations because the smaller slices and numerical factors have changed.

Step 1: use the zero values on sets disjoint from A . For every $X \in \binom{R}{6}$, (24) gives

$$0 = f(X) = \sum_{T \in \binom{X}{3}} c_T.$$

These are the kernel equations for $W_{16,6,3}$, with the coefficients indexed by triples in R . Lemma 3.1 applies because $3 \leq \min(6, 16 - 6)$, and hence

$$c_T = 0 \quad \left(T \in \binom{R}{3} \right). \quad (29)$$

Step 2: use the zero values on sets meeting A in two elements. For distinct $i, j \in A$ and distinct $u, v \in R$, define

$$a_i(\{u, v\}) = c_{\{i, u, v\}}, \quad b_{ij}(u) = c_{\{i, j, u\}}.$$

Thus a_i is indexed by unordered pairs in R , while b_{ij} is indexed by elements of R , and $b_{ij} = b_{ji}$. If $Z \in \binom{R}{4}$, the set $\{i, j\} \cup Z$ meets A in exactly two elements, so its value under f is zero. The coefficients of triples contained in Z vanish by (29). The remaining terms give

$$0 = \sum_{\{u, v\} \in \binom{Z}{2}} (a_i(\{u, v\}) + a_j(\{u, v\})) + \sum_{u \in Z} b_{ij}(u). \quad (30)$$

To apply the rank lemma, we rewrite the single-element sum as a pair sum. Each

element of a four-element set belongs to three of its pairs, so

$$\sum_{u \in Z} b_{ij}(u) = \frac{1}{3} \sum_{\{u,v\} \in \binom{Z}{2}} (b_{ij}(u) + b_{ij}(v)).$$

After this substitution, (30) says that the pair coefficient vector

$$\left(a_i(\{u, v\}) + a_j(\{u, v\}) + \frac{1}{3}(b_{ij}(u) + b_{ij}(v)) \right)_{\{u,v\} \in \binom{R}{2}}$$

belongs to the kernel of $W_{16,4,2}$. Since $2 \leq \min(4, 16 - 4)$, Lemma 3.1 forces every entry of this vector to be zero. Thus

$$a_i(\{u, v\}) + a_j(\{u, v\}) = -\frac{1}{3}(b_{ij}(u) + b_{ij}(v)) \quad (31)$$

for all distinct $i, j \in A$ and distinct $u, v \in R$.

Step 3: apply the linear support bound to sets meeting A in one element. Fix $i \in A$, and define

$$g_i(Z) = f(\{i\} \cup Z) \quad \left(Z \in \binom{R}{5} \right).$$

The map $Z \mapsto \{i\} \cup Z$ sends distinct inputs to distinct six-element sets, so

$$|\text{supp } g_i| \leq |\text{supp } f| = 969.$$

By (29), all triples contained entirely in Z have coefficient zero. Therefore

$$g_i(Z) = \sum_{\{u,v\} \in \binom{Z}{2}} a_i(\{u, v\}).$$

To use Lemma 6.3, we want to express each pair coefficient as a sum of terms depending on single elements. Equation (31) gives such an expression for $a_i + a_j$; we isolate a_i by combining three pairwise sums. For this fixed i , choose distinct $j, \ell \in A \setminus \{i\}$ and keep them fixed as u, v vary. Then

$$\begin{aligned} 2a_i(\{u, v\}) &= (a_i(\{u, v\}) + a_j(\{u, v\})) + (a_i(\{u, v\}) + a_\ell(\{u, v\})) \\ &\quad - (a_j(\{u, v\}) + a_\ell(\{u, v\})). \end{aligned}$$

Substituting (31) into these three sums gives the desired decomposition if we define

$$p_i(u) = -\frac{1}{6}(b_{ij}(u) + b_{i\ell}(u) - b_{j\ell}(u)),$$

Then

$$a_i(\{u, v\}) = p_i(u) + p_i(v) \quad \left(\{u, v\} \in \binom{R}{2} \right). \quad (32)$$

The notation p_i suppresses the fixed choices of j, ℓ ; for each i , one arbitrary choice is sufficient to obtain this decomposition. Consequently,

$$g_i(Z) = \sum_{\{u, v\} \in \binom{Z}{2}} (p_i(u) + p_i(v)) = 4 \sum_{u \in Z} p_i(u). \quad (33)$$

Indeed, every element of a five-element set belongs to four of its pairs. Thus g_i is linear in the indicators of the elements of R . If g_i were nonzero, Lemma 6.3 would give $|\text{supp } g_i| \geq 1001$, contradicting $|\text{supp } g_i| \leq 969$. Hence g_i is identically zero for every $i \in A$.

Step 4: show that all coefficients outside A vanish and recover the star. Our remaining goal is to prove

$$c_T = 0 \quad \text{whenever } T \not\subseteq A. \quad (34)$$

As in the ordinary $q = 3$ proof, this would finish the classification. Indeed, for $T \in \binom{A}{3}$, choose $Z \in \binom{R}{3}$. The set $T \cup Z$ contains exactly one triple lying entirely in A , namely T . Under (34), its coefficient sum would reduce to

$$c_T = f(T \cup Z) \in \{0, 1\}.$$

Together with (28), this would force exactly one triple coefficient to equal 1 and all the others to equal 0. It remains to prove (34).

For each $i \in A$, the conclusion $g_i = 0$ and (33) imply

$$\sum_{u \in Z} p_i(u) = 0 \quad \left(Z \in \binom{R}{5} \right).$$

These are the kernel equations for $W_{16,5,1}$. Lemma 3.1 applies because $1 \leq \min(5, 16 - 5)$, so $p_i(u) = 0$ for every $u \in R$. Equation (32) now gives $a_i(\{u, v\}) = 0$ for every pair in R .

Substituting these zero values into (31) gives

$$b_{ij}(u) + b_{ij}(v) = 0 \quad \left(\{u, v\} \in \binom{R}{2} \right)$$

for every distinct $i, j \in A$. These are the kernel equations for $W_{16,2,1}$. Since $1 \leq \min(2, 16 - 2)$, Lemma 3.1 gives $b_{ij}(u) = 0$ for every $u \in R$.

Step 1 shows that $c_T = 0$ for triples entirely in R . The equations $a_i = 0$ and $b_{ij} = 0$ show that $c_T = 0$ for triples having exactly one or exactly two elements in A , respectively. These are all triples not contained in A , so (34) holds. The coefficient argument above

therefore gives $\mathbf{c} = \mathbf{e}_{T_0}$ for a triple $T_0 \subseteq A$, and hence

$$\mathbf{f} = W_{22,6,3} \mathbf{e}_{T_0} = \mathbf{b}_{T_0}.$$

Thus $f = b_{T_0}$. The uniqueness of the triple coefficient vector also makes T_0 unique. $\square$

6.3 Extension to balanced rainbow slices

We now allow all 22 color classes to have an arbitrary common size $a \geq 1$. The restriction argument from Subsection 5.3 uses only an ordinary support bound, an ordinary equality classification, and Lemma 5.4. Proposition 6.1 and Theorem 6.4 provide the two ordinary results for $(m, k, t) = (22, 6, 3)$. We give the resulting argument with these parameters.

Proof of Theorem 2.3. Let S have 22 color classes, each of size $a \geq 1$, and let $f \in U_3^S(6)$ be nonzero and Boolean. Label the symbols in the i th color class by $(i, 1), \dots, (i, a)$, and write

$$f(X) = \sum_{T \in \binom{X}{3}} c_T \quad (X \in \mathcal{R}_6(S)). \quad (35)$$

To apply the ordinary results, choose one symbol from each color class. For $z \in [a]^{22}$, put

$$V_z = \{(1, z_1), \dots, (22, z_{22})\}, \quad f_z = f|_{\binom{V_z}{6}}.$$

Every subset of V_z is rainbow, so f_z is a Boolean function in $U_3(6)$ on the ordinary slice $\binom{V_z}{6}$. Its triple coefficients are $(c_T)_{T \in \binom{V_z}{3}}$.

To control the support sizes of these restrictions, define

$$h(z) = \frac{|\text{supp } f_z|}{969} = \frac{1}{969} \sum_{X \in \binom{V_z}{6}} f(X) = \sum_{T \in \binom{V_z}{3}} c_T. \quad (36)$$

Indeed, Booleanity gives the middle equality, and each triple in V_z belongs to $\binom{19}{3} = 969$ six-element subsets, giving the last equality. Equivalently,

$$h(z) = \sum_{1 \leq i < j < \ell \leq 22} c_{\{(i, z_i), (j, z_j), (\ell, z_\ell)\}}.$$

Each summand depends on at most three coordinates. Moreover, any $X \in \text{supp } f$ extends to some V_z , and then $f_z(X) = 1$, so h is nonzero. Lemma 5.4 therefore gives $|\text{supp } h| \geq a^{19}$.

The desired support bound follows from

$$a^{19} \leq |\text{supp } h| \leq \sum_{z \in [a]^{22}} h(z) = \frac{a^{16}}{969} |\text{supp } f|. \quad (37)$$

For the middle inequality, Proposition 6.1 says that each f_z is either zero or has support at least 969. Thus $h(z) = 0$ or $h(z) \geq 1$. For the last equality, each rainbow six-element set X is contained in exactly a^{16} sets V_z : its symbols prescribe six coordinates of z , and the remaining sixteen coordinates are arbitrary. Summing $|\text{supp } f_z|$ over z therefore counts every member of $\text{supp } f$ exactly a^{16} times. Dividing (37) by $a^{16}/969$ gives $|\text{supp } f| \geq 969a^3$.

Suppose now that $|\text{supp } f| = 969a^3$. Summing (35) over $\mathcal{R}_6(S)$ gives

$$|\text{supp } f| = 969a^3 \sum_{T \in \mathcal{R}_3(S)} c_T, \quad \text{so} \quad \sum_{T \in \mathcal{R}_3(S)} c_T = 1. \quad (38)$$

Indeed, every rainbow triple belongs to $\binom{19}{3}a^3 = 969a^3$ rainbow six-element sets. To prove that f is a star indicator, it therefore suffices to show that every c_T belongs to $\{0, 1\}$.

Under the equality assumption, both ends of (37) equal a^{19} , so all inequalities in that chain are equalities. Since every nonzero value of h is at least 1, the equality $|\text{supp } h| = \sum_z h(z)$ forces $h(z) \in \{0, 1\}$ for every z . Consequently, each f_z is either zero or has support size 969. By Theorem 6.4, every nonzero restriction f_z is an ordinary triple-star indicator.

To recover the individual coefficients, fix z and write $\mathbf{f}_z = W_{22,6,3}\mathbf{c}_z$, where $\mathbf{c}_z = (c_T)_{T \in \binom{V_z}{3}}$ and the matrix has rows and columns indexed by six-element subsets and triples of V_z . Lemma 3.1 applies because $3 \leq \min(6, 22 - 6)$. Thus $\mathbf{c}_z$ is uniquely determined: it is zero when $f_z = 0$, and it is a standard basis vector when f_z is a star indicator. In particular, $c_T \in \{0, 1\}$ for every triple contained in V_z .

Every rainbow triple is contained in some V_z , by choosing its symbols in its three colors and any symbols in the other colors. Hence $c_T \in \{0, 1\}$ for every $T \in \mathcal{R}_3(S)$. Together with (38), this forces exactly one coefficient $c_{T_0} = 1$ and all other coefficients to be zero. Equation (35) now gives $f = b_{T_0}$.

Conversely, every rainbow triple-star indicator is Boolean, belongs to $U_3^S(6)$, and has support size $\binom{19}{3}a^3 = 969a^3$ by (1). This completes the proof of Theorem 2.3. $\square$

References

- [1] Yaokun Wu, *Forbidden intersections and minimum supports of low-degree Boolean functions on ordinary and rainbow Johnson slices*, manuscript. <https://math.sjtu.edu.cn/faculty/ykwu/data/Paper/Johnson26.pdf>.
- [2] A. E. Brouwer, *The Witt designs, Golay codes and Mathieu groups*, online notes. <https://aeb.win.tue.nl/2WF02/Witt.pdf>.